

T&S HW3

Donation Activity Labeler



Motivation & Harms



Screaming From Under
Children's Dreams.
My Children Are Hungry & The World is Silent.
We are Without a Home, Food, or Medicine 💔.
... Sick Child, Ali ❤️ 🙏
I'm hungry, weak, and can't afford food.
Our children are in the hospital because of hunger.
Each day is harder than the last.
Please, help us 🙏 😭 💔
Even \$1 could mean life for us.
... are going through a real famine
can't even find our daily food 😭 💔 Please
URGENTLY We are begging you from Gaza. Famine will kill us. Our
dream has become to eat. We crave food. We have not eaten
month and a half. We are in dire need of your donations 🙏 🇵🇸
buy food, link don@tions in bio

Technical Framework

Module	1 Donation Intent Classifier	2 Endpoint Extractor	3 Endpoint / Account Verifier	4 Label Assembler
Input	Post text	1) Post text; 2) Embed images (for potential payment QR code); 3) Faceted and embedded payment links	1) Extracted payment endpoints (output from 2); 2) Posting profile	Output of 1 - 3
Goal	1) Detect if a post is donation related; 2) If yes, send it to Endpoint Extractor.	1) Extract any exists payment endpoint from the post; 2) Classify the posts by payment mechanism	1) Classify if the account posting is official or verified; 2) Classify if the payment endpoint is trusted or organizational.	Get the final scam risk labels and prepare for the final output.
Output	pred_donation_related: {yes/no}	pred_contains_payment_mechanism: {yes/no}, pred_payment_mechanism: {...}	pred_verified_domain: {yes/domain/no}, Pred_verified_type: {official/none}	pred_scam: {no/unsure/unknown}

Efficiency

Timing: 133 seconds for **152** samples ~ **0.875s** per sample

```
python policy_proposal_labeler2.py 3.07s user 0.30s system 2% cpu 2:13.24 total
```

Given that we are running mainly text matching and similarity techniques, the labeller is a lightweight and computationally efficient solution to quickly give users signals about the post

Test Dataset Overview

- **Scale:** 163 entries in total (152 posts available until 11/25)
- **Sampling Technique:** Keyword Search (e.g., help out, give money, donate) + Automated Pull (random pull + official charity-related account)
- **Data Schema:**
 - 6 manual labels, where **donation_related**, **contains_payment_mechanism**, and **scam** serve as **targets**, while the other manual labels are designed to help understand the *context*.

Column	donation_related	contains_payment_mechanism	scam	domain	verified_org	Urgent_language
Input	yes	payment link	no	cryptocurrency	yes	yes
		payment handle		war		
		fundraising website		poverty		
		qr code		education		
	no	no	unsure	politics	no	no
		no		disaster relief		
		no		animal		
		no		other		

- **Data Distribution** (based on current available posts):
 - Donation_related: **94 “yes”** v.s. **69 “no”** (include edge cases)
 - Contains_payment_mechanism: **71 “yes”** (60 website + 3 payment link + 5 handles + 3 QR code) v.s. **81 “no”**
 - Scam: **43 “unsure”** v.s. **109 “no”**

Test & Evaluation

- **Accuracy by Task**

Task	Correct	Wrong	Accuracy
donation_related?	139	13	91.54%
has_payment?	139	13	91.54%
payment_mechanism?	139	13	91.54%
scam? ("unknown" ignored)	123	22	84.83%

- **Accuracy by Payment Mechanism**

Payment Mechanism	Correct	Wrong	Accuracy
Fundraising Website	52	8	85.25%
Payment Links	2	1	66.67%
Payment Handles	4	1	80.00%
QR Code	0	3	0.00%
No	81	0	100.00%

- **Precision, Recall, and F1 Score for Intent & Payment Detection**

Donation Related	Pred Related	Pred Not Related	Recall
Actual Related	83	2	<u>0.98</u>
Actual Not Related	11	56	0.84
Precision	<u>0.88</u>	0.97	F1 = <u>0.93</u>

Has Payment	Pred Yes	Pred No	Recall
Actual Yes	58	13	<u>0.82</u>
Actual No	0	81	1
Precision	<u>1</u>	0.86	F1 = <u>0.89</u>

- **Precision, Recall, and F1 Score for Scam Detection**

- We applied a *3-tier* label for scam risk:
 - **"No"**: Trustable Donation
 - **"Unknown"**: Risk level unknown
 - **"Unsure"**: Potential high risk
- Take **"No"** and **"Unsure"** for metrics calculation:
 - 95.4% coverage ratio (145/152)

Scam	Pred Not Scam	Pred Risky	Recall
Actual Not Scam	101	2	0.98
Actual Risky	20	22	<u>0.52</u>
precision	0.84	<u>0.92</u>	F1 = <u>0.67</u>

Stress Test, Edge Case & Risks: Module 1

The Baltimore Krampus Market was such a blast!

- ❄️ Caught up with all of the amazing vendors from last year.
- ❄️ A bunch of Kidd Krampus plushies found their forever homes.
- ❄️ As a cherry on top, some Plushpanions were purchased just to

donate to Toys for Tots! ❤️

#krampus #plush #mothman



PLS DONT SKIP !!

KHALED STILL NEEDS DONOS TO AFFORD A TENT FOR HIS FAMILY,
CLOTHES & BLANKETS

!! ONLY 45\$ IS LEFT TO COMPLETE THE GOAL !!
IT M U S T BE DONE BY FRIDAY TO SECURE THE TENT

PLS DONATE IF YOU CAN & HELP AS MUCH & HOWEVER YOU CAN
PLEASE SHARE TOO

#mutualaid #gaza #palestine #helpsky 💖💰🍉



- Keywords hit but not actually related -> Need context information -> LLM
- Avoid detections by spelling errors -> Fuzzy matching + Similarity Threshold (Implemented)

Stress Test, Edge Case & Risks: Module 2

MATCH UPDATE:

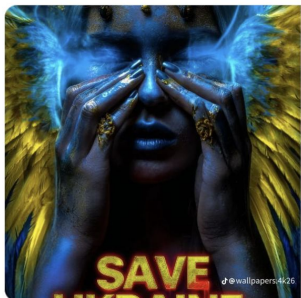
This path to the match was particularly long and difficult.
And time is running out...

\$ 760 / \$ 1,000
\$ 240 to go

Please, boost, RT and donate if you can. We really need your help especially now... 🙏❤️🙏❤️🙏❤️



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PLEASE BOOST, REPOST
DONATE...



Please share & donate to help the Nashwan family survive Israel's continuing genocide in Gaza, which has left Palestinians homeless and starving in brutal winter conditions:



Rebuilding dreams the dreams of my children

Mohammad Nashwan: The Story of a Father Who Dreams of Rebuilding His Life for His Children

© chuffed.org

PLS DONATE IF YOU CAN & HELP AS MUCH & HOWEVER YOU CAN
PLEASE SHARE TOO

#mutualaid #gaza #palestine #helpsky 🙏❤️



URGENT



105/150€ (tent, clothes,
blankets, ect)



Needed ASAP

- Different format of payment endpoints (Implemented, but need improvement with QR Code)
 - In text urls & payment handles + embedded urls + facets urls + qr code payment links

Stress Test, Edge Case & Risks: Module 2 & 3

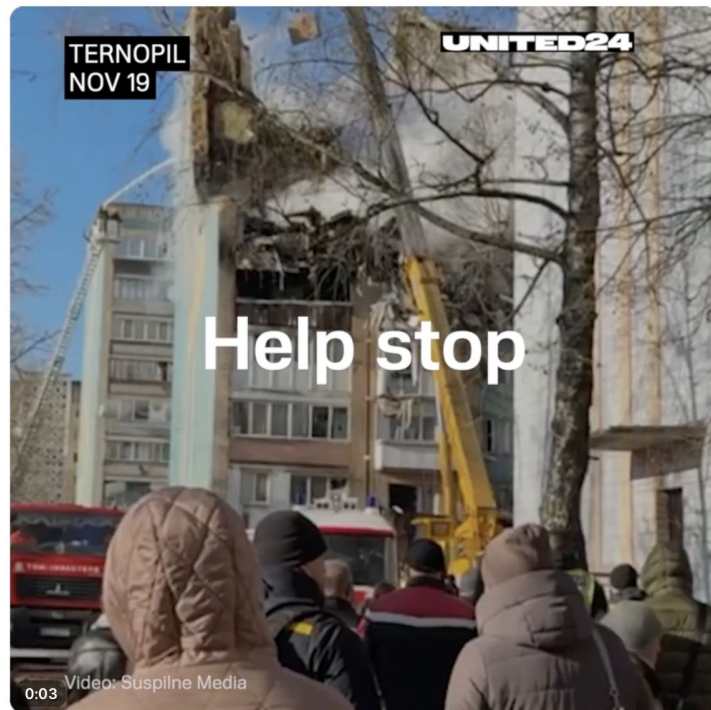
- To determine if a URL is payment related:
 - Check if the domain has been recorded in .JSON
 - Check if the path / subdomain includes keywords
- Missing actual risky sites (**False Negatives**)
 - [Web Risk API](#)
 - LLM based Context Information

If you have the means, consider donating to Transanta. It is set up to provide trans youth, often those who are homeless or in unsupportive environments, with gifts during the holidays. You can read letters and registries to help someone directly, or donate money to ensure everyone gets something.



This is why you have to support Ukraine.

Every donation helps:
u24.gov.ua/nafo-dark-ni...



Iterations, Alternatives & Tradeoffs

- “Deciding” if the post is a scam:
 - Technically challenging and difficult for us to make a highly accurate decision
 - More reasonable to provide the user with signals about the post, so that they can make a more informed decision before donating
- Trained models vs Text based classification:
 - Lack of data to train models on donation intent
 - Reached out to “Pirates of Charity” researchers to get curated donation posts for future work
- LLMs or API based solutions:
 - Budget & Efficiency

Ethical Implications & Future Works

False Negatives – fail to catch scams

False Positives – legitimate fundraisers labeled as "risky"

Account engagement history, Post history consistency

Check if trusted accounts are resharing/endorsing the fundraiser

Adjust risk scoring by cause type dynamically: Lower thresholds for disaster or war appeals as urgent language and new accounts are expected; Higher thresholds for routine causes like poverty, education

Ethical Implications & Future Works

Global Equity and Cultural Context

Multilingual and Multicultural Expansion

Collaborate with charity verification organizations to maintain up-to-date databases of legitimate organizations around the world

Improve Verification Pathways for Grassroots Organizers